KUNWAR ANSHUMAAN SINGH

+91 9118011449 | kunwarsingh2425@gmail.com | LinkedIn | GitHub | Portfolio
Lucknow, Uttar Pradesh, India

RESEARCH EXPERIENCE

Research Intern – CSIR-Indian Institute of Toxicology Research (IITR) *8 June 2025 – 8 September 2025*
Lucknow, Uttar Pradesh, India

- Investigated inhaled particulate deposition in the respiratory system and assisted in vivo mouse inhalation studies, handling dosing, exposure, and tissue collection
- Applied computational tools (MPPD, ConsExpo) and plethysmography to link exposure data with inflammatory and toxicological outcomes

Thesis: *Fate of Inhaled Li-ion Battery Fire Smoke Particles in the Airway using MPPD Modeling*

→ Calculated MMAD (0.66 µm) and GSD (1.8) from MOUDI cascade impactor data on LIB fire smoke, used as inputs for MPPD v3.04 lung dosimetry simulations

→ Predicted regional airway deposition across adult/elderly and male/female models under an oronasal breathing scenario

→ Combined SEM and ion chromatography data on the smoke extracts to assess occupational inhalation risk, under Dr. Jyotsna Singh

Research Intern – Central Ayurveda Research Institute (CARI) *15 January 2025 – 15 February 2025*
Jhansi, Uttar Pradesh, India

- Assessed microbial load of herbal raw materials for contamination and host-microbe interaction risk via CFU-based quantification (TVC, TAMC, TYMC)

Thesis: *Evaluation of Microbial Load of Herbal Raw Materials and Finished Product*

→ Prepared sterile culture media (SCA, SDA) and ran serial dilution (10^{-1} to 10^{-5}) with pour/spread plating for CFU enumeration

→ Confirmed suspected contaminants like Staphylococcus aureus via selective media streaking

→ Compiled QC reports linking contamination sources to processing conditions, under Dr. Hemant Soni

Project: Comparative WGS Analysis of *Saccharomyces cerevisiae* YJM451 *2026*

GitHub: github.com/kunwaranshumaan/yeast-genome-analysis-pipeline

- Developed a reproducible WGS variant-calling pipeline for S. cerevisiae YJM451 using FastQC, fastp, Bowtie2, SAMtools and BCFtools.
- Performed SNP/indel annotation and raw–trimmed variant comparison using Python, identifying 99.50% Jaccard similarity across call sets.

EDUCATION

Master of Science in Biotechnology *July 2024 – July 2026*
Bundelkhand University, Jhansi, India

Bachelor of Science (Honors) in Biotechnology *November 2021 – June 2024*
Bundelkhand University, Jhansi, India

PUBLICATIONS

- **Singh KA**, Dubey A, Tisato V, Choudhary S, Gemmati D, Singh S, Tiwari H, Verma A, Gupta P, Singh AV and Gautam V (2026) Nanoparticle-induced interference with common in vitro assays: a systematic review. Front. Bioeng. Biotechnol. 14:1933980. doi: 10.3389/fbioe.2026.1933980
- Choudhary S., Dubey A., **Singh A.**, Zamboni P., Gupta N., Singh R., et al. Engineering the microenvironment: advanced biomaterials for humanized in vitro immunotoxicology and carcinogenicity assessment. Explor BioMat-X. 2025;2:101351. https://doi.org/10.37349/ebmx.2025.101351

CONFERENCES

Attended Association of Microbiologists of India (AMI) Conference, Bundelkhand University, Jhansi *December 2023*

CERTIFICATES

Next Generation Sequencing Technologies: Data Analysis and Applications | NPTEL, IIT Kharagpur | 12-Week Certification Course *3 November 2026*

Instrumentation-powered Research Methodology - Flow Cytometry | RCB & Waters Biosciences, *22 July 2026*

Elements of AI | University of Helsinki & MinnaLearn (2 ECTS) *1 August 2026*

Advancing Discovery with Live Cell Imaging and Analysis | Labroots *29 July 2026*

International Workshop on Molecular Dynamics Simulation | Quaxon Bio & IT Solutions *December 2025*

International Virtual Workshop: Molecular Phylogenetics in Bioscience Research - From Theory to Practice | Quaxon Bio & IT Solutions *August 2025*

Proteintech Webinar: Understanding Neurodevelopment via Brain Organoids *17 October 2024*

Proteintech Webinar: How to Characterize an Antibody *24 October 2024*

AWARDS

Summer Internship Project – 1st position in Master's thesis presentation | Bundelkhand University, Jhansi *27 February 2026*

Silver Medal – Runner Up in Scientific paper writing & documentation | Bundelkhand University, Jhansi *12 December 2024*

First Prize – National Science Day Symposium (Vigyan Diwas) | Bundelkhand University, Jhansi *February 2024*

SKILLS

- **Molecular & Cellular Biology:** Aseptic technique, culture media prep, CFU quantification, basic PCR/ELISA, SEM, Photospectrometry, Flow Cytometry, Chromatography, Fluorescence microscopy, live-cell imaging and time-lapse analysis
- **Toxicology & Exposure Modeling:** MPPD, ConsExpo, aerosol dosimetry, MMAD/GSD
- **Bioinformatics & Computational Biology:** NGS Data Analysis Fundamentals, NCBI Databases and Tools, SwissADME, Molecular Docking (SwissDock), Molecular Dynamics Simulation (NAMD, VMD), PyMOL, Chimera, Machine Learning Fundamentals, AI Applications in Biological Research
- **NGS & Genomic Data Analysis:** Illumina WGS, FASTQ quality assessment, adapter trimming, reference alignment, SAM/BAM processing, coverage analysis, SNP/indel calling, VCF normalization and variant annotation
- **Programming / Data:** Python, R (basic), Anaconda, Bash scripting, Excel, Word, Git and GitHub
- **Nanomaterial exposure:** eNanomapper, Styrene modeling
- **Documentation:** Scientific writing, data interpretation
- **Core Competencies:** Collaboration, interdisciplinary work

RECOMMENDATIONS

Dr. Ajay Vikram Singh
Senior Scientist
Dept. of Chemical and Product Safety
German Federal Institute for Risk Assessment (BfR)
Max-Dohrn-Strasse 8-10, 10589, Berlin, Germany
Phone: +49-301-8412-1263
Email: Ajay-Vikram.Singh@bfr.bund.de

Dr. S.V. Singh
Assistant Professor
Dept. of Biotechnology
Bundelkhand University, Jhansi, UP, India, 284128
Phone: +91 80040 08896
Email: drsvs@bujhansi.ac.in